THE FIBRE PRODUCT OF TWO FINITE TYPE ALGEBRAS OVER A NOETHERIAN RING IS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Artin-Tate lemma on finiteness of algebras generated by a finite module. the author cites Algebra, Lemma algebra-lemma-Artin-Tate at proof line 361. Supplies the criterion that converts the finiteness of one algebra over the fibre product into finite generation.
- Fibre product of algebras and the exact sequence relating it to the two factors. used at proof lines 339-350 where the commutative diagram with exact rows is displayed. Supplies the injectivity and the module structure used at the end.
- Finite generation of an algebra as a module over a subalgebra. used at proof lines 350-360. Provides the generators 1, x_1 to x_n of one factor over the fibre product.

Source reference. Original lemma 00IT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 691–691. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00IS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 690–690. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $A \to B$ and $C \to B$ be R-algebra maps. Assume*

(1) *R is Noetherian,*

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(2) A, B, C *are of finite type over* R,
(3) $A \to B$ *is surjective, and*
(4) B *is finite over* C.

Then $A \times_B C$ *is of finite type over* R.

Proof. Set $D = A \times_B C$. There is a commutative diagram

$$\begin{array}{ccccccccc} 0 & \leftarrow & B & \leftarrow & A & \leftarrow & I & \leftarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \leftarrow & C & \leftarrow & D & \leftarrow & I & \leftarrow & 0 \end{array}$$

with exact rows. Choose $y_1, \ldots, y_n \in B$ which are generators for B as a C-module. Choose $x_i \in A$ mapping to y_i. Then $1, x_1, \ldots, x_n$ are generators for A as a D-module. The map $D \to A \times C$ is injective, and the ring $A \times C$ is finite as a D-module (because it is the direct sum of the finite D-modules A and C). Hence the lemma follows from the Artin-Tate lemma (Algebra, Lemma 00IS). $\square$